PROFESSIONAL EDUCATION



SUMMER 2024

DESIGNING EFFICIENT DEEP LEARNING SYSTEMS

DATES	DURATION	FORMAT	FEE	CEUS
July 8–9, 2024	2 Days	On Campus or Live Virtual	$2,500	1.3

Discover how to build and utilize deep learning systems

Deep learning is widely used for many artificial intelligence (AI) applications including computer vision, speech recognition, robotics, etc. While deep learning delivers state-of-the-art accuracy on many AI tasks, it requires high computational complexity. Accordingly, designing efficient hardware systems to support deep learning is an important step towards enabling its wide deployment, particularly for embedded applications such as mobile, Internet of Things (IOT), and drones.

This course aims to provide a comprehensive tutorial and survey about the recent advances towards enabling the efficient processing of deep learning. Specifically, it will provide an overview of deep learning, discuss various hardware platforms and architectures that support deep learning, and highlight key trends in recent efficient processing techniques that reduce the cost of computation for deep learning either solely via hardware design changes or via joint hardware design and network algorithm changes. It will also summarize various development resources that can enable researchers and practitioners to quickly get started on deep learning design, and highlight important benchmarking metrics and design considerations that should be used for evaluating the rapidly growing number of deep learning hardware designs, optionally including algorithmic co-design, being proposed in academia and industry.

EARN A PROFESSIONAL CERTIFICATE

This course may be taken individually or as part of the ***Professional Certificate Program in Machine Learning & Artificial Intelligence***.

WHO SHOULD ATTEND

This course is designed for research scientists, engineers, developers, project managers, startups and investors/venture capitalists who work with or develop artificial intelligence for hardware and systems, as well as mobile or embedded applications:

- For **project managers and investors/venture capitalists** whose work involves assessing the viability or potential impact of a deep learning system and selecting a research direction or acquisition, this course aims to provide an overview of the recent trends as well as methods to assess the technical benefits and drawbacks of each approach or solution based on a comprehensive set of metrics.
- For **research scientists and engineers** whose work involves designing and building deep learning systems, this course aims to provide an overview of the various state-of-the-art techniques that are being used to address the challenges of building efficient deep learning systems.
- For **startups and developers** whose work involves developing deep learning algorithms and solutions for embedded applications and systems, this course aims to provide the insights necessary to select the best platform for your goals and needs. It will also highlight techniques that can be applied at the algorithm level to improve the energy-efficiency and speed of your proposed solution.

professional.mit.edu/dls

LEARNING OUTCOMES

- Understand the basics of deep learning, how it is applied to various applications, and how it is processed on various platforms
- Outline the key design considerations for deep learning systems
- Be able to evaluate different deep learning implementations with benchmarks and comparison metrics
- Understand the strengths and weakness of various hardware architectures and platforms
- Be able to assess the utility of various design techniques for efficient processing for deep learning
- Understand and evaluate recent implementation trends and opportunities in deep learning systems

LEAD INSTRUCTOR

Vivienne Sze



Vivienne Sze is an associate professor in MIT's Department of Electrical Engineering and Computer Science and leads the Research Lab of Electronics' Energy-Efficient Multimedia Systems research group. Her group works on computing systems that enable energy-efficient machine learning, computer vision, and video compression/processing for a wide range of applications, including autonomous navigation, digital health, and the internet of things. She is widely recognized for her leading work in these areas and has received many awards, including faculty awards from Google, Facebook, and Qualcomm, the Symposium on VLSI Circuits Best Student Paper Award, the IEEE Custom Integrated Circuits Conference Outstanding Invited Paper Award, and the IEEE Micro Top Picks Award. As a member of the Joint Collaborative Team on Video Coding, she received the Primetime Engineering Emmy Award for the development of the High-Efficiency Video Coding video compression standard. She is a co-author of the book entitled "Efficient Processing of Deep Neural Networks".

Teaching Methodology

The certificate curriculum is grounded in the spirit of MIT's motto, "Mens et Manus," or "mind and hand," which combines theoretical instruction with hands-on methods of discovery.



INTERACTIVE LECTURES



HANDS-ON EXERCISES



GROUP PROJECTS



SUPPLEMENTAL RESOURCES

PROFESSIONAL EDUCATION

ABOUT MIT PROFESSIONAL EDUCATION

For 70 years, MIT Professional Education has been providing technical professionals worldwide a gateway to renowned MIT research, knowledge, and expertise, through advanced education programs designed specifically for them. In addition to industry-focused, two-to-five-day live virtual and on-campus courses through Short Programs, MIT Professional Education offers professionals the opportunity to take online and blended learning courses through Digital Plus Programs, attend courses abroad through International Programs, enroll in regular MIT academic courses through the Advanced Study Program, or attend Custom Programs designed specifically for their companies. For more information, please visit **professional.mit.edu**.